

Project :- Imphal Airport			
Component Name : Duct Insulation			
Make Offered :			
Date of Submission :			
Sr. No	Tender Specification	Compliance/ Deviation	Comments from Client/Consult
1	Duct Insulation		
1	a)Insulation material shall be Closed Cell Elastomeric Nitrile Rubber with 7 mils factory laminated glass cloth on one side. b)Thermal conductivity of insulation material shall not exceed 0.035 W/ (m.K) at mean temperature of 0°C as per EN 12667. c)Moisture Diffusion Resistance Factor or'μ' value of insulation material shall be minimum 7,000 without any external barrier as per EN 12086. d)The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and also passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990. e)The insulation material shall have fire performance of VO, HB as per UL 94, 1996. f)The insulation material shall be FM (Factory Mutual), USA Approved. g)Density of insulation material shall be between 40 to 60 Kg /m3. h)The insulation material shall be dust and fiber free. i)The insulation material shall withstand maximum surface temperature of +85 Deg.C and minimum surface temperature of 0 Deg.C as per EN 14706. j)Glass Cloth: Weight shall be 205 +/- 10 grams per square meter, Tensile Strength shall be 275 +/- 25 Kg / 50 mm (minimum) and Thickness shall be 0.18 mm / 7 mil. Glass Cloth shall be UV resistant for artificial weathering. k)The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential of Zero.		Comply
	Thickness of the insulation shall be as specified for the individual application.		
	The insulation material shall be installed as per manufacturer's recommendation.		
1	Duct Acoustic Lining- Open Cell Nitrile Rubber		
	Acoustic insulation material shall be Open Cell Engineered Nitrile Rubber foam.		
	aT) he acoustic insulation material shall be dust and fibre free. bT) he acoustic insulation material shall have good sound absorption, damping and barrier properties. cT) he density of acoustic insulation material shall be within 140-180 Kg/m3. dT) he acoustic insulation material shall have antimicrobial product protection, and shall pass Fungi / Moulds Resistance as per DIN EN ISO 846method And Bacterial Resistance as per DIN EN ISO 846 method C, from an independent testing agency. eT) he acoustic insulation material shall have a thermal conductivity not exceeding 0.047 W/ (m.K)@ 20 Deg. Cas per EN 12667. fT) he acoustic insulation material shall withstand maximum surface temperature of +85 Deg.C and minimum surface temperature of -20 Deg.C as per EN 14706. gT) he material shall conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & UL 94 (HB, V-0) in accordance to UL 94, 1996. hT) he acoustic insulation material shall pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 without any laminate or covering (section 12.7), from an independent accredited testing agency. iT) hickness of the insulation shall be as specified for the individual application.		Comply
1	Acoustic Lining of Mechanical Rooms-Open Cell Nitrile Rubbe		

	Air handling unit rooms shall be provided with acoustic lining. Material shall be processed from Nitrile rubber open cell foam.	
	he material should be fiber free. The density of the same shall be 140-180 Kg/m3. b)The material should have thermal conductivity not exceeding 0.047 W/Mk at an average temperature of 20° C. c)The maximum surface temperature of material shall withstand is 85° C and minimum temp shall be -20° C. Thickness shall be as specified. d)The material should conform to class 1 rating for surface spread of flame as per BS 476 Part 7 & HBF, HF1 & FH 2 in accordance with UL 94, 1996. Thickness, if not specified, shall be considered as 20mm specified for the individual application and with noise absorption proprieties (NRG 0.60) as per IS: 8225 I ISO 354 / ASTM423C. e)Insulation should have antimicrobial product protection, and should pass Fungi Resistance as per DIN EN ISO 846 Method A for fungal resistance and Method C for bacteria resistance. The insulation should pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 (section 12. 7). f)Surface shall be cleaned and two coats of adhesive recommended by the manufacturer should be applied on the walls. The foam sheets should be cut to required size and a layer of adhesive should also be applied to it. When it is tack dry it is stuck to the walls I ceiling. g)All longitudinal and transfer joint shall be covered with 22 gauge 50 mm wide GI strip with screws. h)Acoustic lining of walls shall be terminated approximately 15 cm above the finished floor to prevent damage to insulation due to accidental water-logging in plant/AHU rooms.	Comply